



Molecular dynamics simulation of the microscopic mechanisms of the dissolution, diffusion and aggregation processes for waxy crystals in crude oil mixtures

Yifan Gan, Qinglin Cheng*, Zhihua Wang, Jinwei Yang, Wei Sun, Yang Liu

Key Lab of Ministry of Education for Enhancing the Oil and Gas Recovery Ratio, Northeast Petroleum University, Daqing, 163318, China

ARTICLE INFO

Keywords:

Molecular dynamics simulation
Waxy crude oil
Phase transition
Wax gelation
Coupling effect

ABSTRACT

To ensure the safe and economic operation of a waxy crude oil production process, the microdynamic mechanism and thermodynamic characteristics of the wax gelation process need to be revealed at nanoscale. The molecular dynamics model was established to characterize the phase transition and gelation behavior of waxy molecules in a multiphase system (including oil, asphaltene and water). The relative error between the simulated results and experimental data measured by Dutour et al. (2002) was less than 5%. Under the coupling effect of different operation parameters, the molecular dynamics simulation was employed. The simulated results showed that the spherical paraffin crystals underwent the processes of dissolution, diffusion and aggregation. After which waxy cluster crystals with larger amount but smaller volume were formed, which would be deposited on the inner wall under the concentration gradient. The influence mechanisms of different operating parameters on wax gelation were analyzed. And it was found that the increase in temperature and water cut decrease the wax precipitation, while the increase in pressure enhance the wax precipitation rate. Furthermore, by means of hydrogen bonding and the effect of similar dissolution, water and asphaltene molecules also affect the wax precipitation process at the molecular scale. The investigations in this study provide theoretical support for the paraffin removal and control in a waxy crude oil production system.

1. Introduction

Waxy crude oil is the complex mixture composed of large number of hydrocarbons. The hydrocarbon components include saturated hydrocarbons (n-alkanes, isoalkanes, naphthenes), aromatic hydrocarbons, colloids, asphaltenes, etc. The paraffin in waxy crude oil is the mixture composed of n-alkanes (the molecular weight is high, the carbon atom number is more than 16) and a small amount of isoparaffins, naphthenes, aromatic hydrocarbons, the total carbon number distribution is generally between 16 and 70 (Singh et al., 2000; Quan et al., 2015). With the waxy crude oil extracted from reservoirs, depending on the operating condition such as temperature and pressure changes, the dissolved waxy molecules in oil flow will precipitate into solids through phase transition. Then, the wax molecules will diffuse toward the inner wall of pipeline due to the radial concentration gradient, which causes immense economic losses to important areas of oil industry production, such as reservoir development, crude oil exploitation, long-length pipeline transportation and oil and gas gathering pipeline transportation (Liu and Chen, 1999; Cao et al., 2019). For the wax deposition existing

in the above important areas of oil industry production, it is a complex phenomenon involving series of thermo-hydraulic coupling process, the operation parameters and the composition of crude oil components may have a great effect on wax deposition rate. So, it is necessary to have a thorough understanding of the detailed process and internal mechanism of wax deposition, which helps the engineers make the accurate formulation of relevant flow assurance schemes. In addition, the fundamental understanding of the interaction rules, thermophysical properties and dynamic characteristics for wax deposition process is also necessary to carry out.

To explore the mechanism of precipitation process for waxy molecules in crude oil multiphase systems, experimental analysis and molecular dynamics (MD) simulation have been widely applied by researchers. In terms of microscopic experimental studies, Moussa Kane et al. (Kané et al., 2003) adopted the transmission electron microscope to observe the morphological difference under different cooling conditions (the observation scale is 1 μm) and found that the lamellar and layered crystals existed in waxy crude oil during static cooling; furthermore, when the amount of wax precipitation is larger, the lamellar

* Corresponding author.

E-mail address: chengqinglin7212@163.com (Q. Cheng).

<https://doi.org/10.1016/j.petrol.2019.04.059>

Received 30 November 2018; Received in revised form 9 April 2019; Accepted 16 April 2019

Available online 19 April 2019

0920-4105/ © 2019 Elsevier B.V. All rights reserved.

and layered crystals were cross-linked and aggregated into clusters. Jaber Taheri-Shakib et al., 2018a, 2018b adopted the scanning electron microscope to observe the waxy crystallization behavior and found that the paraffinic crystals consisted of a network of large tubular crystals similar to rice grains (the observing scale is 10 μm), revealing that the dispersion degree, average size and morphological structure of wax crystals mainly depend on the molecular structure, element composition and asphaltene content. Yao et al., 2016, 2018 modified an original hydrophilic nano-clay by octadecyltrimethyl ammonium chloride (OTAC) and polyoctadecylacrylate (POA), and used microscopic to observe the images (10 μm) of crude oil undoped/doped with POA and POA/clay nanocomposite PPD. The results showed that the addition of POA/clay nanocomposite PPD to Changqing waxy crude oil improves the rheological properties while elevating the WAT (wax appearance temperatures) of the oil. To sum up, the relevant microscopic experimental studies were still in micron scale, it was difficult to observe the inherent complexity of wax gelation process such as microscopic properties and structural information.

In terms of macroscopic experimental researches. The DSC thermal analysis technology was adopted by Piroozian A et al. (Piroozian et al., 2016), the WAT (wax appearance temperatures) was obtained, but this method can't reveal the detailed process of wax gelation from a microscopic perspective. In addition, this method needs the actual oil samples and the differential scanning calorimetry, the research cost is higher. Jeramie J. Adams et al. (2018) used experimental methods of centrifuge to compare the effectiveness of a commercially available wax inhibitor (WIA) to a newly developed wax inhibitor (WIEP) using a highly waxy Wyoming crude oil, the results showed that the newly developed WIEP additive was significantly more effective at reducing the amount of ambient temperature wax crystallites in the crude oil. But it mainly centrifuged the paraffin to test the inhibitor's effect, no description of wax precipitation process was involved. Singh et al. (2000) conducted the laboratory flow loop experiments. The gelation, diffusion, precipitation and counterdiffusion processes of waxy molecules in deposited gel were studied. They concluded that the precipitation and then diffusion are the primary drivers of wax deposition. For the subprocess of gelation during wax precipitation, it was accompanied by the dissolution, diffusion, and aggregation process, which were the important links in wax precipitation. However, the micro-dynamic mechanism and thermodynamic characteristics of these processes have not been revealed in their work.

In summary, the relevant experimental studies were still performed at the micron scale, and the subprocess of gelation during wax precipitation had not been revealed from a microscopic perspective. Thus, it is difficult to grasp the inherent complexity of wax gelation process, such as microscopic properties and structural information.

The MD simulation method can characterize the microscopic behavior and internal mechanism of materials at nanoscale, which could not be characterized by theoretical analysis and experimental observation method. In addition, the molecular motion, structural variation and bonding properties can be further studied. Therefore, for the waxy crude oil multiphase system, the microdynamic mechanism and thermodynamic characteristics of phase transition and gelation process can be revealed from a microscopic perspective by MD simulation. Which could deepen the engineers' fundamental understanding for the mechanism of waxy crude oil precipitation process, and provide reliable theoretical guidance for the development of waxy crude oil industrial production.

At present, the relevant MD simulation studies on waxy crude oil mainly focus on the inhibition effect of pour point depressant on wax deposition (Duffy and Rodger, 2002; Zhang et al., 2008; Duffy et al., 2004), while research on the wax gelation behavior and related thermodynamic properties is still in its infancy. Chen X et al. (Chen et al., 2017) established the MD model to describe the microscopic characteristics of waxy crude oil transportation process under the magnetic field treatment; the influence mechanism of magnetization on density,

viscosity and aggregation for waxy molecules was also analyzed. Samieadel A et al. (Samieadel et al., 2018) adopted the MD method to investigate the effect of paraffin on molecular conformation and packing in aged asphalt binder; the results of this study show that the waxy molecules can promote dimerization of oxidized asphaltene molecules while suppressing nanoaggregates formation. Nonetheless, the mechanism of phase transition and gelation behavior for waxy crystals in crude oil multiphase system at molecular scale is unclear, especially for the multiphase system containing water and asphaltene molecules. Moreover, the influence mechanism of different operation parameters on wax gelation process has not been revealed, so it is necessary to carry out the relevant research.

In this study, the MD model was established to characterize the phase transition and gelation behavior of waxy molecules in multiphase systems (including oil, asphaltene and water) for the first time. Furthermore, considering the interaction between water molecules, asphaltene molecules and waxy molecules, the density, solubility parameter, self-diffusion coefficient, radial distribution function and changing ratio of energies were statistically calculated under different relaxation temperatures, relaxation pressures and water cuts. Based on this, the microscopic mechanism of dissolution, diffusion and aggregation processes for waxy crystals in multiphase systems was revealed, and the influence mechanism of different operation parameters for the phase transition and gelation processes was further analyzed. The investigations in this study can provide theoretical support for safe and economical operation of waxy crude oil production processes.

2. MD simulation methodology

2.1. System model

2.1.1. The physical properties of Daqing waxy crude oil

In this paper, the waxy crude oil in Daqing oilfield was selected (Liu et al., 2012); a multiphase system mainly including oil, paraffin, asphaltene and water was employed, and the inversion point was 65.2% (the water cut corresponds to the transition point from W/O emulsion to O/W emulsion). The WAT was 345.15 K. The peak wax precipitation point was 315.15 K. The minimum temperature at which the oil acts as a Newtonian fluid was 315.15 K. The oil pour point was 310.15 K. The weight percent of each component in multiphase system were shown in Table 1.

2.1.2. The molecular models

In the field of MD simulation for waxy crude oil, the oil, wax and asphaltene components were always simplified by corresponding molecules. If we didn't simplify the relevant components, various kinds of molecules would make the simulation process more difficult. For example, in Chen's work (Chen et al., 2017), for the light oil, eight types of hydrocarbons including alkanes (144 hexane, 132 heptane, 156 octane, 180 nonane, 96 cyclohexane, 156 cycloheptane) and aromatics (156 toluene, 60 benzene) molecules were used to simplified the oil component. For the paraffin component, 40 n-hexacosane ($\text{C}_{26}\text{H}_{54}$) and 40 n-octatriacontane ($\text{C}_{38}\text{H}_{78}$) molecules were chosen to simplified. In Wu's work (Wu et al., 2005), the normal alkane of $\text{C}_{18}\text{H}_{38}$ was used to simplified the wax crystal. In Samieadel's work (Samieadel et al., 2017), the Martin-Martinez's model (Martín-Martínez et al., 2015) was used to simplified the asphaltene, the $\text{n-C}_{11}\text{H}_{24}$ was used to simplified the paraffin. In Jang's work (Jang et al., 2007), the n-C_7 was used as the oil

Table 1
wt percentage and molecular number of each component.

Component	Oil	Wax	Asphaltene	H ₂ O
Weight percentage	63.44 wt%	23.25 wt%	8.31 wt%	10 wt%
Molecular number	1500	207	49	532

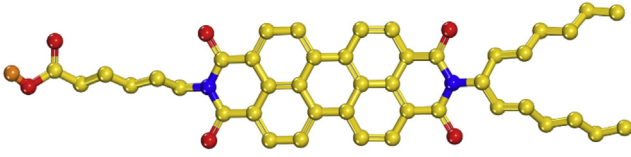


Fig. 1. Molecular structure model of asphaltene.

solvent and n-C₃₂ was used as the wax compound of MD modeling. Therefore, in this paper, we use the method of component simplification to deal with the complex components of waxy crude oil.

The molecular model of each component was constructed in this paper. For light oil, we adopted the Tang's model (the dodecane (C₁₂) was employed as the model of oil molecules) (Tang et al., 2018). To represent paraffin molecules, which are high molecular weight alkanes composed of more than 16 carbon atoms. We adopted the Dutour's model (the mixture of C₂₈H₅₈ and C₃₆H₇₄ molecules were adopted to represent the paraffin molecules) to simplify the paraffin component (Dutour et al., 2002). Asphaltene was the component with the largest average molecular weight, and its heteroatom (oxygen, nitrogen, etc.) was high in content and strong in polarity; thus, the molecular model adopted in this study is shown in Fig. 1 (Gao et al., 2014). All MD simulations in this study were performed using Materials Studio (Accelrys, Inc.).

2.1.3. The construction of the simulation models

To explore the microscopic mechanism of dissolution, diffusion and aggregation processes for waxy crystal in multiphase system. The system model for MD simulation was built. The modeling steps can be summarized as follows: Step 1: the spherical cluster with relative large bulk of waxy crystals (the mixture of C₂₈H₅₈ and C₃₆H₇₄) was built (shown in Fig. 2b), and the diameter of the cluster was 25 Å. Step 2: an amorphous cell system consists of oil molecules (dodecane), asphaltene molecules and water molecules was built by AC (amorphous cell) method, with the simulation box size of 40 Å × 40 Å × 40 Å (X × Y × Z). Step 3: the spherical clusters of wax crystals were put into the amorphous cell system, removing the overlapping parts in amorphous cell. Thus, the simulation system model was obtained, as shown in Fig. 2.

The molecular number of each component was determined according to the mass fraction. The detailed component information of

the selected waxy crude oil multiphase system was presented in Table 1.

2.2. Force field model

The force field model for describing the MD process between wax crystals and crude oil mixtures was established. To make the calculation results more accurate, the established force field model fully considered the intramolecular interaction (including the bond stretching, bond angle bending and dihedral angle torsion) and the intermolecular interaction (the short-range interaction adopted the Lennard-Jones 12-6 potential function and the electrostatic interaction adopted the Coulomb potential function). The complete force field model was as follows (Chen et al., 2017; Jorgensen and Tiradorives, 1996):

$$E(r^N) = E_{\text{bond-stretch}} + E_{\text{angles}} + E_{\text{dihedrals}} + E_{\text{non-bonded}} \quad (1)$$

$$E_{\text{bond-stretch}} = \sum_{\text{bonds-stretch}} K_r (r - r_0)^2 \quad (2)$$

$$E_{\text{angles}} = \sum_{\text{angles}} K_\theta (\theta - \theta_0)^2 \quad (3)$$

$$E_{\text{dihedrals}} = \sum_{\text{dihedrals}} \left(\frac{V_1}{2} [1 + \cos(\phi - \phi_1)] + \frac{V_2}{2} [1 - \cos(2\phi - \phi_2)] + \frac{V_3}{2} [1 + \cos(3\phi - \phi_3)] + \frac{V_4}{2} [1 - \cos(4\phi - \phi_4)] \right) \quad (4)$$

$$E_{\text{non-bonded}} = \sum_{i>j} f_{ij} \left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{C_{ij}}{r_{ij}^6} + \frac{q_i q_j e^2}{4\pi\epsilon_0 r_{ij}} \right) \quad (5)$$

by combining the rules $A_{ij} = \sqrt{A_{ii}A_{jj}}$ and $C_{ij} = \sqrt{C_{ii}C_{jj}}$.

The potential function parameters, Lennard-Jones 12-6 and Coulomb, for the particles of wax, oil, asphaltene and water are listed in Table 2.

2.3. Governing parameters

2.3.1. The MD simulation algorithms

The temperature control method in simulation system adopted the algorithm of Nose-Hoover thermostat (Morishita, 2010). While the Berendsen method was used for pressure control (Feller et al., 1995). The long-range Coulomb electrostatic interaction among particles

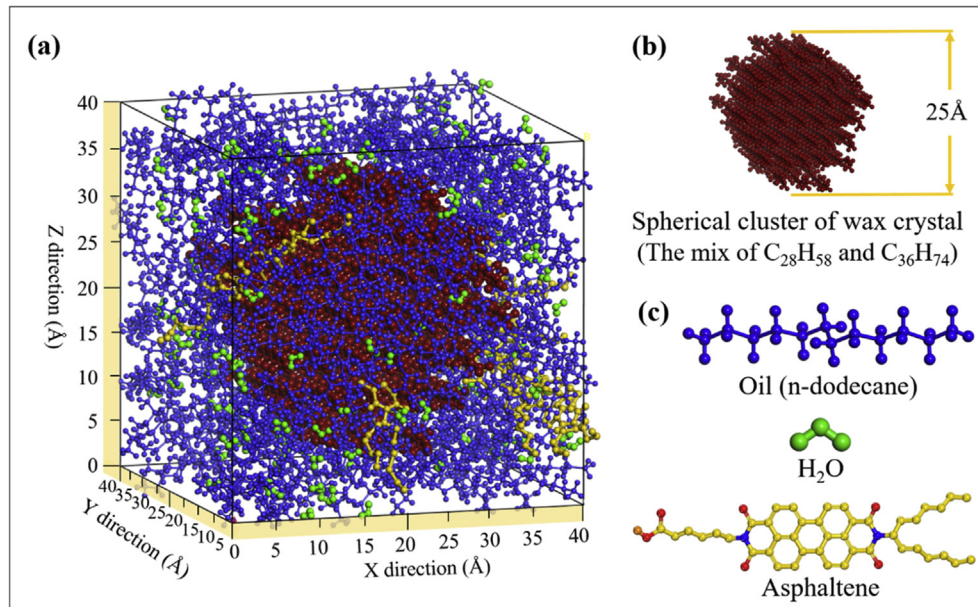


Fig. 2. The system model and details.

Table 2
Parameters used in the potential function.

Type	Molecules	Q/e	$\sigma/\text{\AA}$	$\epsilon/\text{kJ}\cdot\text{mol}^{-1}$
C	Alkanes ^c	0.000	3.20	0.051
H		0.045	2.50	0.030
C ^a	Asphaltenes ^d	0.000	3.20	0.051
H ^a		0.000	2.50	0.030
C ^b		0.223	3.0	0.049
H ^b		0.106	2.5	0.029
O		−0.450	2.31	0.570
N		−0.223	3.32	0.814
H	H ₂ O ^e	0.4238	3.30	0.037
O		−0.8476	2.31	0.650

^a the C and H atoms in Asphaltene's branched alkanes.

^b the C and H atoms in Asphaltene's aromatic.

^c the alkanes in wax molecules and oil molecules (Kahn and Bruice, 2002).

^d (Boek et al., 2009).

^e (Zhang and Duan, 2005).

adopted the algorithm of particle-particle particle-mesh (PPPM), with an accuracy of 10^{-4} . The atom-based algorithm was used for calculating Van der Waals interaction, with a cut-off radius of 20 Å. The initial velocity of each molecule was given by Boltzmann distribution, and the Verlet-leapfrog method was used for solving the motion equation.

2.3.2. The MD simulation steps

The simulation steps can be summarized as follows:

Step 1: The system was energy-minimized using the Smart algorithm and run for 1000 steps, and the time step was set to 1 fs for the complete simulation process.

Step 2: The system was annealed under the NPT (The isothermal-isobaric ensemble, of which the atomic number N, pressure P and temperature T of system were kept unchanged) or NVT (The canonical ensemble, of which the atomic number N, volume V and temperature T of system were kept unchanged) ensemble to eliminate the effect of internal stress among molecules. The system was heated from 323.15 K to 423.15 K, then cooled to 323.15 K; the temperature interval was 20 K and each cycle included five repetitions. The system was annealed for 400 ps in the NPT ensemble and 600 ps in the NVT ensemble, and the total annealing time was 1000 ps. After annealing, the energy and density of system were stable, and the unstable conformations were gradually eliminated.

Step 3: The multiphase system was simulated for 500 ps under the NPT ensemble to maintain the stability of simulation box, with a simulated temperature of 323.15 K and a pressure of 0.1 MPa (the simulation convergence standard was that the energy, temperature, density and box size fluctuated within 5%).

Step 4: The system was simulated for 1000 ps under the NVT ensemble to maintain the external pressure balance of system, with a simulated temperature of 323.15 K (the simulation convergence standard was that the energy and temperature fluctuated within 5%).

Step 5: The system was simulated for 600 ps under the NVE (The microcanonical ensemble, of which the atomic number N, volume V and energy E of system were kept unchanged) ensemble to employ the calculation results (the simulation convergence standard was the same as step 4).

2.3.3. The cross coupling operating conditions of MD simulation

To further analyze the microscopic behavior between wax crystals and multiphase system under different operating conditions. The MD simulation was carried out using the established method, for which the parameters of cross coupling conditions are listed in Table 3.

Table 3
Cross coupling conditions in this study.

Items	Number	Operating conditions
Temperature	1	(T = 363.15 K, P = 0.1 MPa, ϕ = 10%)
	2	(T = 343.15 K, P = 0.1 MPa, ϕ = 10%)
	3	(T = 323.15 K, P = 0.1 MPa, ϕ = 10%)
Pressure	4	(T = 363.15 K, P = 6 MPa, ϕ = 10%)
	5	(T = 363.15 K, P = 3 MPa, ϕ = 10%)
	6	(T = 363.15 K, P = 0.1 MPa, ϕ = 10%)
Water cut	7	(T = 363.15 K, P = 0.1 MPa, ϕ = 30%)
	8	(T = 363.15 K, P = 0.1 MPa, ϕ = 20%)
	9	(T = 363.15 K, P = 0.1 MPa, ϕ = 10%)

3. Results and discussion

3.1. Model validation

The researches of Wang et al. (2015) and Luo et al. (2005) had showed that the simulation methods and calculating results of MD model would be reliable if the relative error between the simulated density and experimental density was within 10%. Therefore, we adopted this method to validate the reliability of the established model. Furthermore, in this paper, for carrying out the MD simulation, the Dutour's model was used to represent the paraffin component. In Dutour's paper (Dutour et al., 2002), the densities of paraffin at different temperatures and pressures were experimentally measured. Therefore, in order to validate the accuracy of the established model, we carried out the comparison of paraffin's density between the Dutour's experimental results and the simulated results.

The pressure and water cut of the system were kept unchanged (pressure 0.1 MPa, water cut 10%). The model established in the second chapter was adopted to simulate the system at different temperatures. As the results obtained from MD simulation were the atomic trajectories under the NVE ensemble, we must use the theory of statistical thermodynamics to transform the simulation results into macroscopic physical quantities; the density of system can be calculated as follows (Zhou et al., 2011):

$$\rho(z) = [N(z - \Delta z/2, z + \Delta z/2)]/(\Delta z \times S) \quad (6)$$

where z is the length of a certain direction, S is the basal surface area, and $N(z - \Delta z/2, z + \Delta z/2)$ is the average number of waxy atoms occurring in the interval of $(z - \Delta z/2, z + \Delta z/2)$ (Δz selected as 0.75 Å in this study).

The densities of waxy molecules at different relaxation temperatures were statistically calculated, and Fig. 3a was obtained. Next, the temperature and water cut of system was kept unchanged (temperature 363.15 K, water cut 10%). The system was simulated under different relaxation pressures, and the densities of wax under different pressures were statistically calculated, as shown in Fig. 3b.

For the actual data used for comparing with the simulation results, the experimental results of Dutour's (Dutour et al., 2002) et al. were selected. But, the number of Dutour's data used for validation was very limited. For the waxy densities in the range of temperature (273.15K–363.15 K) and pressure (45 MPa–65MPa), we combined the ASPEN PLUS software with NIST (National Institute of Standards and Technology) database to obtain them. The comparison results were shown in Fig. 3.

For the relative errors between our simulation results and the experimental results (Xie et al., 2018), we defined the relative errors of δ_s and δ_p as $\frac{\rho_{\text{This Paper's Model}} - \rho_{\text{Actual Experimental Data}}}{\rho_{\text{Actual Experimental Data}}}$. The relative errors under different simulated temperatures and pressures were calculated, which can be seen in Table 4.

From Fig. 3 and Table 4, the simulated results of waxy density at different temperatures and pressures were basically in agreement with the experimental results of Dutour's (Dutour et al., 2002) et al. and

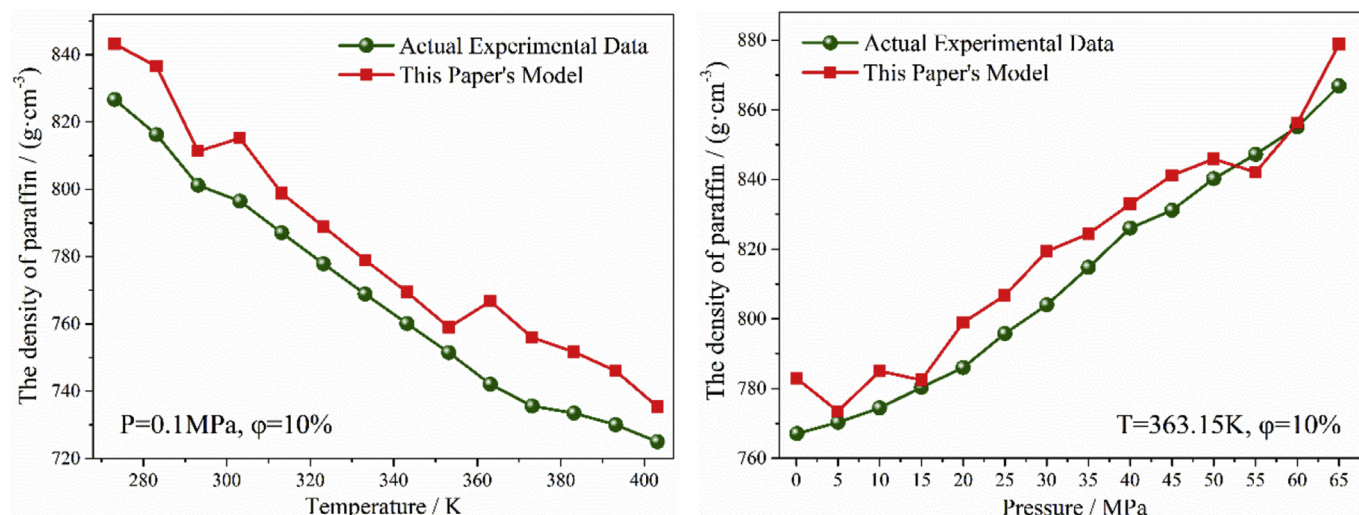


Fig. 3. The density comparison of simulation results and actual experimental results under different temperatures/pressures.

Table 4

The relative error between simulation results and actual experimental results under different temperatures/pressures.

Temperature (K)	$\delta(\%)$	Pressure (MPa)	$\delta(\%)$
273.15	3.9973	0.1	4.0537
283.15	4.4909	5	2.3169
293.15	3.2612	10	3.3700
303.15	4.3454	15	2.1740
313.15	3.4989	20	3.6505
323.15	3.4215	25	3.3772
333.15	3.3115	30	3.9114
343.15	3.2285	35	3.1894
353.15	3.0005	40	2.8415
363.15	4.8181	45	3.2100
373.15	4.7607	50	2.6700
383.15	4.4801	55	-1.9983
393.15	4.2009	60	2.1280
403.15	3.4362	65	3.3883

NIST's. The relative error was less than 5%, which showed that the results simulated by the established MD method were reliable. So, it could be used to study the microscopic behavior of dissolution, diffusion and aggregation for waxy molecules in multiphase system.

3.2. Microscopic behavior of wax gelation in multiphase system

The multiphase system was simulated under condition 1. The instantaneous snapshot of the relative positions for waxy molecules, oil molecules, asphaltene molecules and water molecules were collected, which was shown in Fig. 4.

To investigate the phase transition and gelation processes of waxy crystals in multiphase systems, it is necessary to define the minimum intermolecular distance at which the waxy clusters could be treated as the waxy crystals. In this study, when the centroidal distance between two molecules is less than $1.5 \sigma_{c-c(\text{paraffin})}$ (the value is $4.8 \text{ \AA}$, as given in Table 2), the two molecules can be treated as being in the same crystal cluster (Migdisov et al., 2014).

By calculating the distance between waxy molecules at each stage in Fig. 4, we can conclude that the initial large waxy spherical clusters have been partially dissolved in the multiphase system at 300 ps, most of the crystal structure has disappeared, and only traces of a few crystal cavities can be occasionally found, which indicates that the paraffin began to translate from the initial solid phase to liquid phase. At 900 ps, the structure of each crystal cavity disappears completely, and the centroidal distances between waxy molecules are all larger than $4.8 \text{ \AA}$,

which indicates that the waxy molecules have been completely dissolved and diffused to different parts of simulation box, and the waxy molecules are irregular and begin to distort partially. At 1200 ps and 1500 ps, the formation tendency of waxy clusters is obvious, and the partial distances between waxy molecules are less than $4.8 \text{ \AA}$. At 1800 ps, some small clusters have been formed, indicating that the phase transition had occurred. At 2100 ps, the system has achieved the final equilibrium, and 5 waxy clusters with smaller volume are formed, with the waxy molecules in clusters of 8, 6, 7, 15 and 12 respectively, which was shown in Fig. 4. Furthermore, the molecular arrangement is more regular, and the intermolecular distance is smaller, which indicates that the waxy clusters develop into waxy crystals, and the paraffin translates from liquid phase to solid phase. In addition, the water molecules aggregate and partially bind to waxy molecules, the potential reason for which is the existence of hydrogen bonds. In summary, the MD behavior between the large initial waxy cluster and the multiphase system can be summarized as follows: the larger initial wax crystals dissolve by phase transition at first, then the dissolved waxy molecules diffuse continuously and form the irregular distribution. Finally, some smaller wax crystals around the box are formed by phase transition and aggregation, which leads to a certain concentration gradient of system, and the wax crystals will precipitate accordingly. Furthermore, the multiphase system was simulated under condition 2 – condition 9 respectively, and the same microbehaviors of waxy molecules in the multiphase system were found, similar to condition 1.

3.3. Influence mechanism of different working conditions on waxy density distribution

In Wang's work (Wang et al., 2015), they proposed the theory that the occurrence of "solid-like" substances in fluid can be determined by the occurrence of each peak on the density distribution curve. Using the statistical calculation method proposed in formula (6), we calculated the waxy density distributions in the X, Y and Z directions under condition 1 (baseline), condition 3 (low temperature), condition 4 (high pressure) and condition 7 (high water cut). Next, the spatial distribution of specific particles or surfaces was studied. The density distributions are shown in Fig. 5.

From Fig. 5, the density distributions in different directions are uneven with various sizes of peaks. As wax aggregation would lead to a sudden increase in density, the appearance of the peak can be regarded as the formation of a waxy molecular cluster. The system relaxation was conducted under condition 1 using MD simulation. The waxy density

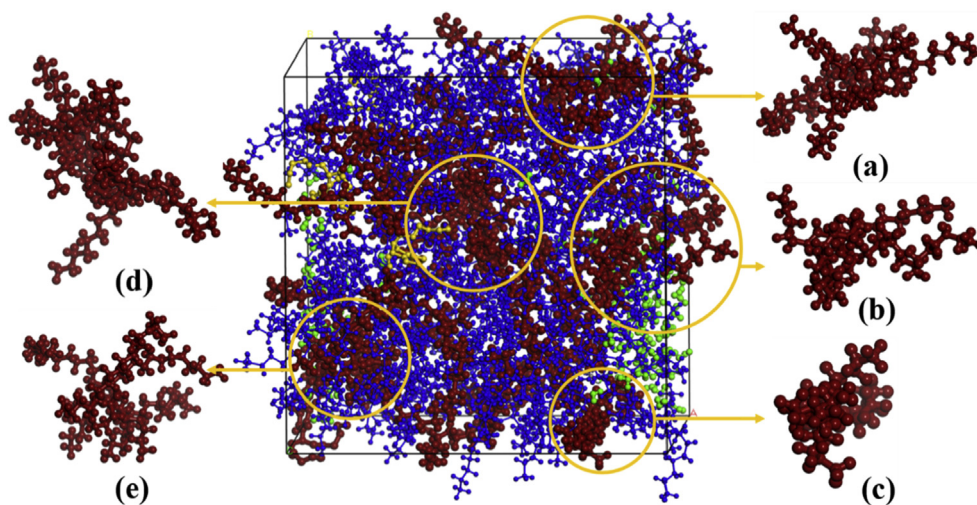


Fig. 4. The snapshots of multiphase system at 2100 ps (the final configuration) for condition 1.

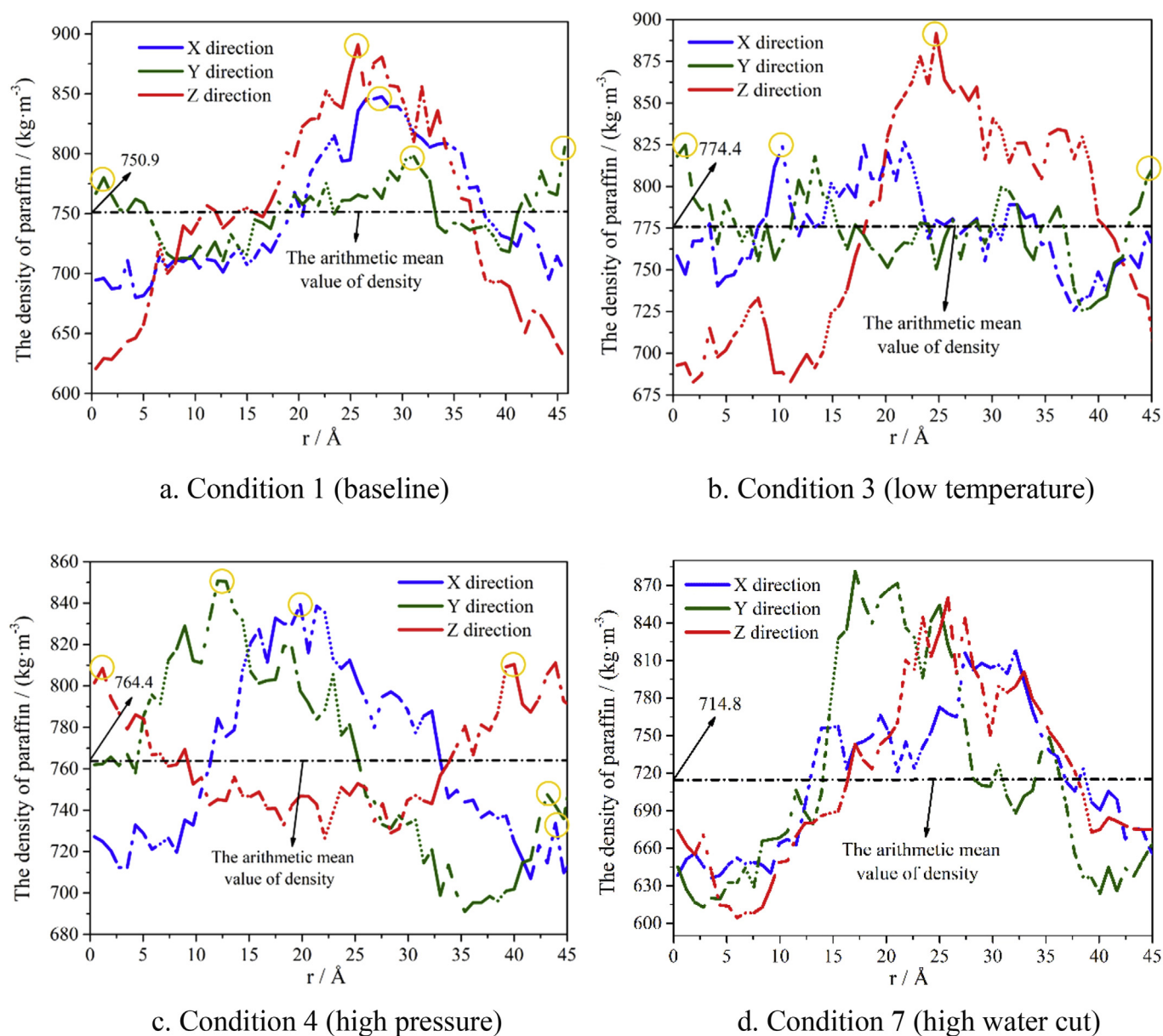


Fig. 5. Density distribution of waxy molecules in different direction.

was statistically calculated (Fig. 5a), and it was observed that five sudden increases of densities appear in the X, Y and Z directions, the appearance of these five peaks could be regarded as the formation of five wax molecular clusters, which corresponds to the conclusion that the system has formed as 5 waxy clusters (obtained in 3.2 section). From the above analysis, we can conclude that the method of “peak of density method can be used to determine the number of waxy clusters” is feasible.

When the temperature decreases from 363.15 K to 323.15 K, there are four peaks from Fig. 5b (lower temperature). This indicates that the decrease of temperature reduces the number of clusters and increases the average density of wax molecules. The reason is that the reduction of temperature decreases the solubility of wax molecules in the multiphase system, and the diffusion ability of wax molecules decreases, the dispersion degree in the unit box is low. Meanwhile, with the decrease of system temperature, the amount of the thermodynamic energy converting to kinetic energy decreases, the movement of molecules weakens. The wax molecules tend to aggregate into larger clusters, which raises the degree of cluster compactness. Therefore, in unit volume, the number of waxy molecules increases, the number of clusters decreases to four, the wax molecules in unit volume box exist in the form of several larger clusters, the corresponding density increases.

When the pressure rises from 0.1 MPa to 6 MPa (Fig. 5c), the dissolution and diffusion ability increase, and the waxy molecules increase in unit volume, which leads to the enhancement of average paraffin density accordingly. Meanwhile, the increase in pressure reduces the distance between waxy molecules, which enhances the intermolecular interaction between them. The ability of aggregation increases accordingly. Therefore, the number of waxy clusters increases to six.

When the water cut rises from 10% to 30% (Fig. 5d), the dissolution ability increases. But because the water molecules have the electrostatic shielding effect on waxy molecules, the diffusion ability reduces, and the number of waxy molecules in unit volume decreases, which leads to the decrease in waxy density and the number of waxy clusters. In addition, the increase in water cut increases the amount of hydrogen bonds between water and wax; the hydrogen bonds will restrict the motion ability of waxy molecules, and intermolecular aggregation between waxy molecules is more likely to occur. Thus, the volume of each cluster increases, which increases the value of each peak.

3.4. Dissolution behavior of waxy molecules in multiphase system

To analyze the dissolution mechanism of waxy crystals (with relatively large volume) in multiphase systems at nanoscale, it is necessary to quantitatively study the solubility parameters δ by MD simulation. The solubility parameter δ can be expressed as the square root of Cohesive Energy Densities (CED) $d_{C,E}$:

$$\delta = d_{C,E}^{1/2} = \left[\frac{\Delta E}{V} \right]^{1/2} = \left[\frac{\Delta H_V - RT}{\bar{V}} \right]^{1/2} \quad (7)$$

Here, ΔE is the cohesive energy, $\bar{V}$ is the molar volume of liquids, ΔH_V is the molar vaporization heat, and RT is the required expansion work when liquid converting to gas. According to the theory of intermiscibility judgement, the closer the two solubility parameters are, the better the blending effect of the two materials will be. Generally, when the D -value (difference value) between the two solubility parameters is less than $0.5 \text{ cal}^{1/2} \text{ cm}^3/2$, the solubility is believed to be better.

In the crude oil multiphase system, the asphaltene contains alkyl side chains, which may increase the dissolution ability of waxy molecules in mixtures. In this study, the relaxation of asphaltene-containing system and asphaltene-free multiphase system under different working conditions was performed; the solubility parameters of paraffin and multiphase system were calculated accordingly, and then the D -value between them was determined, as shown in Fig. 6.

From Fig. 6, with the increase in relaxation temperature, the D -

value of solubility parameters between wax and multiphase system decreases, which indicates that the increase in temperature enhances the dissolution ability. The microscopic mechanism of the above phenomena can be explained as follows: the dissolution between wax and multiphase system is an endothermic process, and the increase in temperature is beneficial to the endothermic reaction. Therefore, the increase in temperature will promote the dissolution ability accordingly, which confirms the validity of the conclusion (obtained in section 3.3) that the dissolution ability will be improved by raising the temperature.

Under the same water cut and temperature, with the increase in relaxation pressure, the dissolution ability will be enhanced. When the pressure further increases, the variation trend will be flat. The reason can be summarized as follows: the waxy molecules have long alkane chains with strong mobility; the increase in pressure will increase their bending, compression and mobility effect, and the internal chemical bonds will be compressed accordingly. Furthermore, the increase in pressure will weaken the distance between wax and other molecules. The above intramolecular and intermolecular effects will jointly enhance the intermolecular interaction between them, and the dissolution ability will be enhanced accordingly. However, when the pressure continuously increases, the bending and compression effect of internal chemical bonds may reach the limit. The distance between wax and other molecules also reaches the limit, so the intermolecular interaction between them may not increase when the pressure continuously increases.

When the relaxation temperature and pressure are constant, the increase in water cut will enhance the dissolution ability. The reason for this may be attributed to the increase in hydrogen bond between O atom in water molecules and H atom in waxy molecules. At present, many studies (Szczęśniak et al., 1993; Raghavendra and Arunan, 2008) have confirmed that the hydrogen bonds can be formed with the structure of $\text{C}-\text{H}\cdots\text{O}$ in alkanes and water molecules. Based on the MD simulation results in this study, the hydrogen bonds in system are calculated. Taking condition 1 as example, after the 2100 ps simulation, the hydrogen bonds have formed between waxy molecules and water molecules, and part of hydrogen bonds snapshots are shown as below.

As seen in Fig. 7, the hydrogen bonds between water molecules have formed, which makes the water molecules present a cage-like structure. In addition, the C-H in alkyl molecules will act as the donor of hydrogen bond, while the O atom in water molecules will act as the acceptor, which forms a hydrogen bond with the structure of $\text{C}-\text{H}\cdots\text{O}$. Therefore, with the increase in water cut, the dissolution ability of a waxy molecule in multiphase systems will be enhanced accordingly.

By comparing the D -value for the solubility parameters of the waxy molecules in asphaltene-containing and asphaltene-free multiphase systems, we can conclude that the dissolution ability would be enhanced if the asphaltene existed. The microscopic mechanism can be explained as follows. The asphaltenes contain alkyl side chains, and the waxy molecules have relative longer alkyl side chains. According to the principle of similar dissolve mutually theory, the waxy molecules are easier to dissolve in a multiphase system composed of oil and asphaltene, which is consistent with the conclusion of reference (Li et al., 2016). In addition, in Kriz's work (Kriz and Andersen, 2004), it was found that the increase of asphaltene concentration could lead to the WAT and yield stress decrease, which indicated that the presence of asphaltene increased the solubility of paraffin in waxy crude oil multiphase system. This is consistent with the conclusion obtained from Fig. 6 in this paper that “The dissolution ability would be enhanced if the asphaltene existed”, the accuracy of the conclusions of the relationship between waxy solubility and asphaltene can be validated.

3.5. Diffusion behavior of waxy molecules in a multiphase system

After the waxy crystals (with relatively large volume) dissolve in a multiphase system, the dissolved waxy molecules will diffuse. To

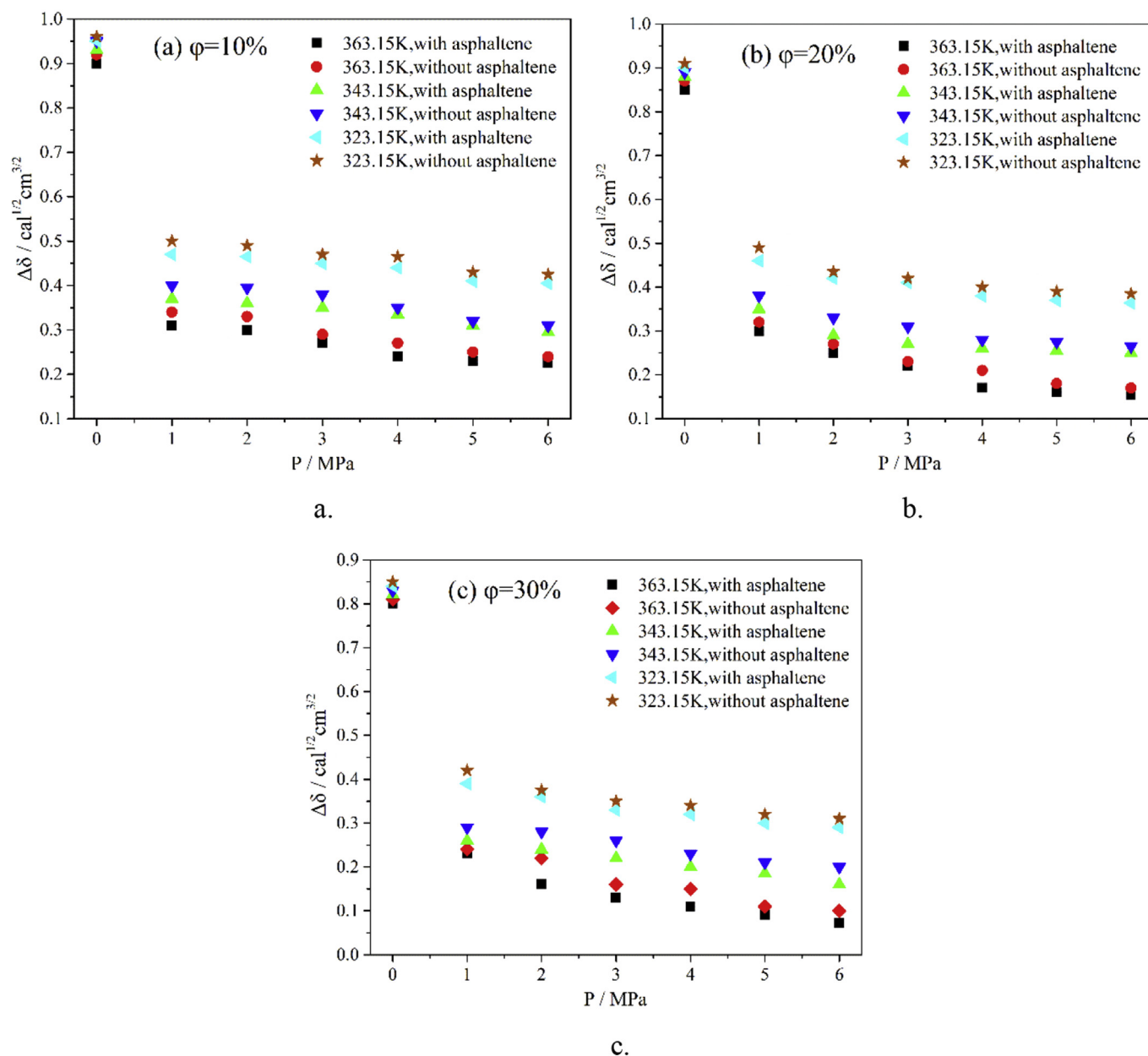


Fig. 6. D-value of solubility parameters versus pressure under different water cut and temperature.

analyze the influence mechanism of different operating parameters on the diffusion behavior for waxy molecules, the self-diffusion coefficients were calculated according to the results of MD simulation. The diffusion coefficient D of waxy molecules can be expressed as $D = m/6$, where m was the slope of Mean Square Displacement (MSD) to time t .

In order to validate the correctness of the simulation results of the established model for diffusion behavior of waxy molecules in water-in-oil emulsion under different water cuts. The established MD model was adopted to simulate the diffusion behavior under the water cut of 0%, 10%, 30%, 50% and 70% respectively. The corresponding diffusion coefficient D was calculated. Among them, when the water cut was 0%, the diffusion coefficient was denoted as D_{single} . Then the D/D_{single} values under different water cut were determined. Next, the relative errors between Zheng's experimental results and our simulated results were calculated (Zheng and Fogler, 2014; Panacharoensawad and Sarica, 2012). The comparison results for simulated values and experimental values of diffusion coefficients at different water cuts were shown in Fig. 8.

From Fig. 8, the relative errors between Zheng's experimental results and our simulated results were less than 10%.

In order to validate the correctness of the simulation results of the established model for diffusion behavior under different temperatures. The established MD model was adopted to simulate the diffusion coefficients of paraffin at different temperatures of 283.15 K, 303.15 K, 323.15 K, 343.15 K and 363.15 K. Correspondingly, under the same temperatures, the diffusion coefficients were calculated using Hayduk's model (Hayduk and Minhas, 1982). Next, the relative errors between Hayduk's model results and our simulated results were calculated.

The comparison results for simulated values and Hayduk's model values of diffusion coefficients at different temperatures were shown in Fig. 9.

From Fig. 9, the relative errors between Hayduk's model results and our simulated results were less than 10%. So, the simulation results of diffusion behavior for waxy molecules in waxy crude oil multiphase system under different water cuts and temperatures were correct. In addition, through literature researches, there were few experimental

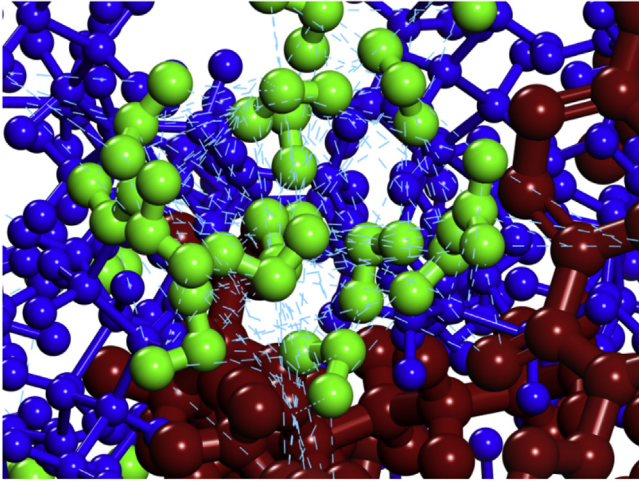


Fig. 7. Snapshots for part of the hydrogen bond in multiphase system.

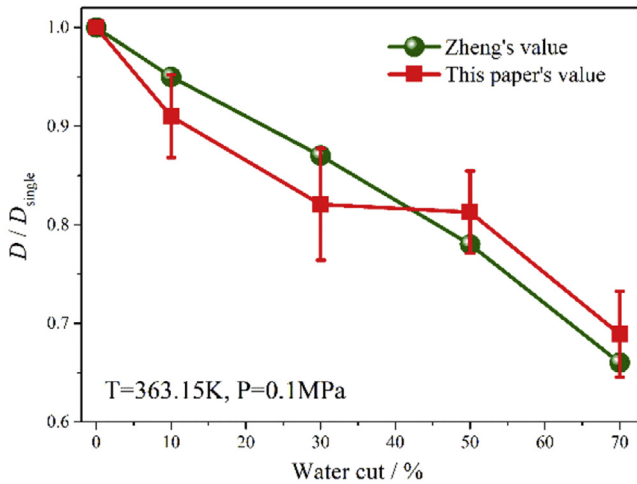


Fig. 8. The diffusion coefficients of the simulated and experimental values at different water cuts.

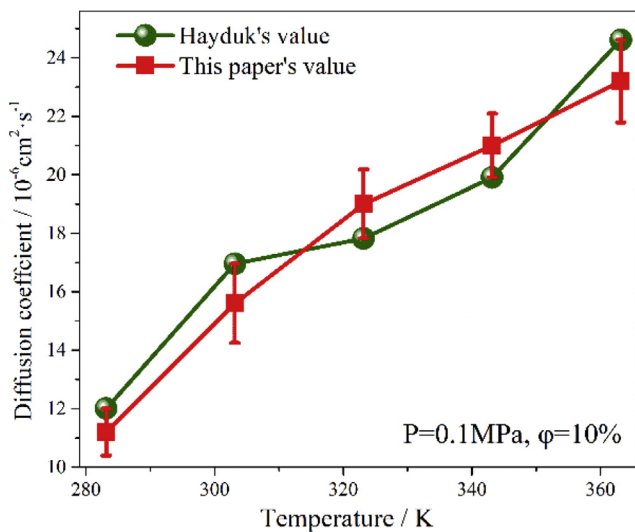


Fig. 9. The diffusion coefficients of the simulated and Hayduk's values at different temperatures.

data on diffusion coefficients of paraffin in waxy crude oil under different pressures. Therefore, we can only firstly compare the diffusion coefficients of waxy molecules at different water cuts and temperatures to illustrate the accuracy of the simulation results for diffusion coefficients. The next, under different temperatures and different water cuts, the relationship between the self-diffusion coefficient D and the relaxation pressure was calculated, as shown in Fig. 10.

From Fig. 10, with the increase in relaxation pressure, the diffusion coefficients of waxy molecules increase continuously. When it reaches a certain pressure, the variation trend slows down gradually. The microscopic mechanism of the above phenomena can be explained as follows: with the increase in relaxation pressure, the mobility of waxy molecules increases, which leads to the increase in diffusion ability. When the pressure increases to a certain value, the binding degree between waxy molecules and other molecules increases, and the diffusion capacity reaches the limit. Thus, the increase rate of self-diffusion coefficient becomes slower. In Cheng's and Liu's work (Liu et al., 2018; Cheng et al., 2017), they proposed that the diffusivity of waxy molecules would increase with the increase of pressure from the point of view of non-equilibrium thermodynamics. The correctness of their conclusion was validated by carrying out the wax deposition experiments using the flow chart of simulation experimental device. Which validated the conclusion of the relationship between diffusivity coefficients and pressure obtained in this paper.

Under the same water cut and pressure and with the increase in the relaxation temperature, the self-diffusion coefficients increase. In addition, the higher the temperature of system, the larger the variation trend of self-diffusion coefficients. The reason is that the thermal movement of waxy molecules increases with the increase in temperature, so the diffusion ability will be enhanced accordingly. Moreover, owing to the cross-coupled effect of temperature and pressure on wax diffusion behavior, the two effects enhance the diffusivity of waxy molecules together (Liu et al., 2018; Gan et al., 2018).

Under the same temperature and pressure, with the increase in water cut, the self-diffusion coefficients of waxy molecules decrease gradually. The reason is that the higher the content of water molecules in unit volume, the higher the collision probability between water and wax, which will hinder the diffusion of waxy molecules.

3.6. Aggregation of waxy molecules in multiphase systems

According to the analysis in section 3.1, after the waxy molecules dissolve and diffuse in a multiphase system, the waxy molecules will aggregate into the waxy crystals with a more regular molecular arrangement. To explore the microscopic mechanism of waxy molecular aggregation process and reveal the influence mechanism of different operating parameters on it, the RDF (radial distribution function) of waxy molecules is proposed in this study. For the RDF of $g_{C_7-C_8}(r)$, the C_7 atom in the waxy molecule is used to represent the central waxy particles and the C_8 atom in the waxy molecule is used to represent other waxy particles; the ratio of the probability for other waxy particles appearing in spherical shell (taking the central waxy particles as the center, with the radius of R , the thickness of dr) following a random distribution can be calculated accordingly. The RDF $g_{C_7-C_8}(r)$ of waxy particles can be expressed as follows:

$$g_{C_7-C_8}(r) = \frac{dN_{C_7-C_8}}{4\rho_{C_8}\pi r_{C_7-C_8}^2 dr} \quad (8)$$

Here, ρ_{C_8} represents the number density of C_8 , $N_{C_7-C_8}$ represents the number density of C_8 atoms surrounding C_7 atoms in the range of r to $r + dr$, and $r_{C_7-C_8}$ represents the coordination shell radius in the range of r to $r + dr$. According to the results of MD simulation, the RDF of each working condition is calculated, shown in Fig. 11.

It can be seen from Fig. 11a that the positions and shapes for the peaks and troughs in RDF curves are identical at different relaxation

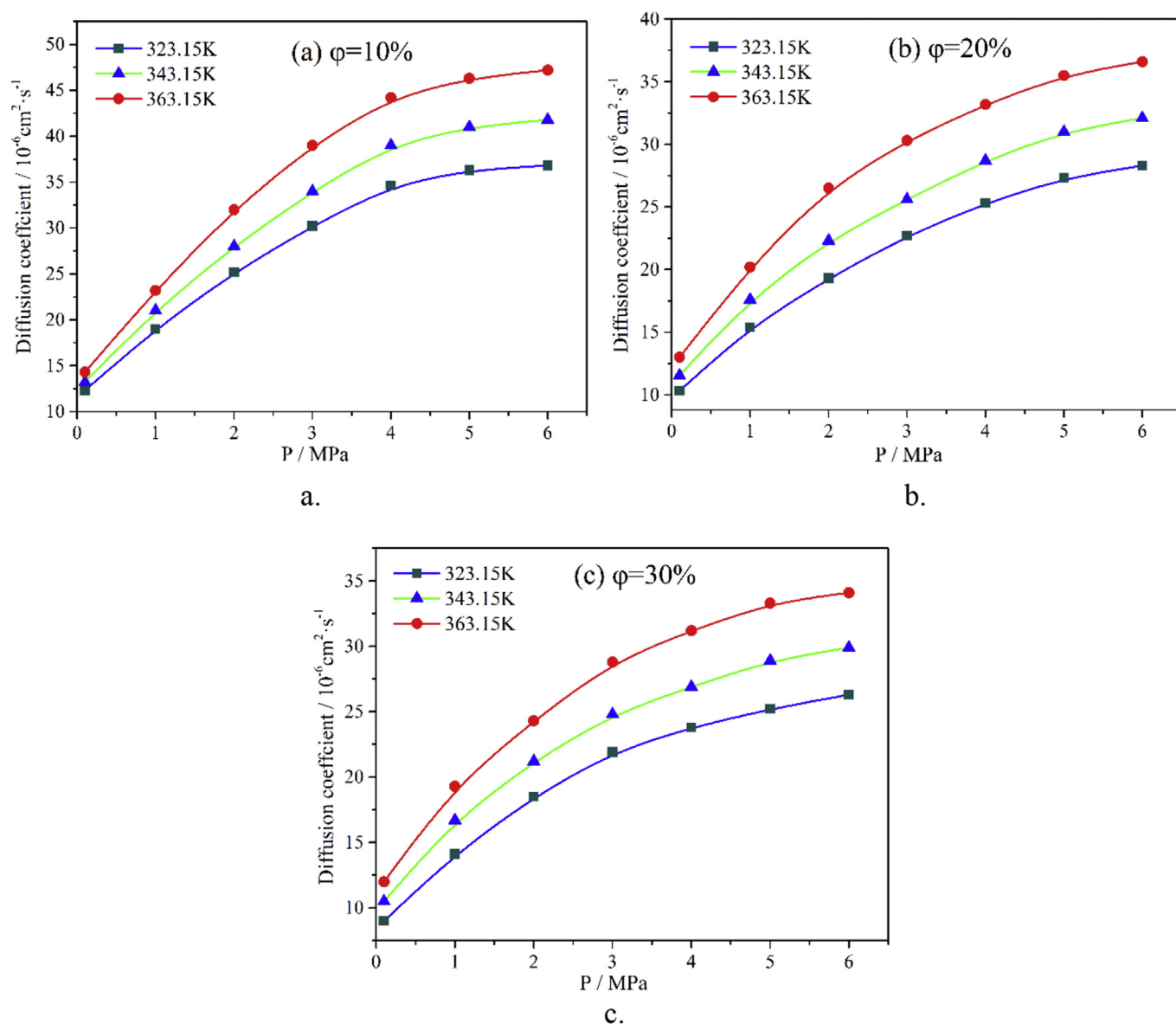


Fig. 10. Self-diffusion coefficients versus pressure under different water cuts and temperatures.

temperatures, but their values are different. The position of the first peak appears at $1.535 \text{ \AA}$, which indicates that the waxy particles aggregate here. In addition, the peak value decreases with the increase in temperature, which is 67.17, 44.72 and 26.31 respectively. The microscopic mechanism of the above phenomena can be explained as follows: with the increase in temperature, the binding energy is released in the form of thermal energy, which leads to the decrease in binding energy between waxy molecules. In addition, the increase in temperature also enhances the mobility of waxy molecules, so the distance between molecules will increase accordingly, which will weaken the intermolecular interaction and electrostatic interaction between waxy molecules, reduce the probability of mutual aggregation, and decrease the value of *RDF*.

From Fig. 11b, with the same temperature (323.15 K) and water cut (10%), the positions and shapes for the peaks and troughs in *RDF* curves are identical at different relaxation pressures, but their values are different. The position of the first peak appears at $1.525 \text{ \AA}$. With the increase in pressure, the *RDF* value is 67.17, 44.72 and 26.31 respectively, which indicates that the higher the pressure is, the higher the aggregation degree of waxy molecules will be. The reason can be explained as follows: the waxy molecules (the mixtures of n-octacosane

and n-hexadecane) have high compressibility and large free space; under high pressure, waxy molecules are more likely to form compact clusters. With the decrease in pressure, the compression degree of waxy molecules decreases, which weakens the aggregation degree of waxy molecules; thus, the size of the formed waxy crystals will be smaller, and the structural distribution will be sparser.

From Fig. 11c, the positions and shapes for the peaks and troughs in *RDF* curves are different at different water cuts. Their values are different too. With the increase in water cut, the *RDF* curve reaches the peaks at 1.525, 1.475, and 1.375, and the value of peaks gradually increase to 43.02, 55.71 and 66.67, respectively, indicating that the increase in water cut makes the clusters appear closer to the central atom and that the aggregation degree is higher. The microscopic mechanism of the above phenomena can be explained as follows: the hydrogen bond formed between water molecules and waxy molecules has a certain restrictive effect on the mobility of waxy molecules. With the increase in water cut, this restrictive effect will be enlarged, and the interaction between waxy molecules will occur more easily; the interaction force will be enhanced accordingly, which leads to the increase in the aggregation degree for waxy molecules. In addition, the maximum water cut corresponding to each working condition in this study

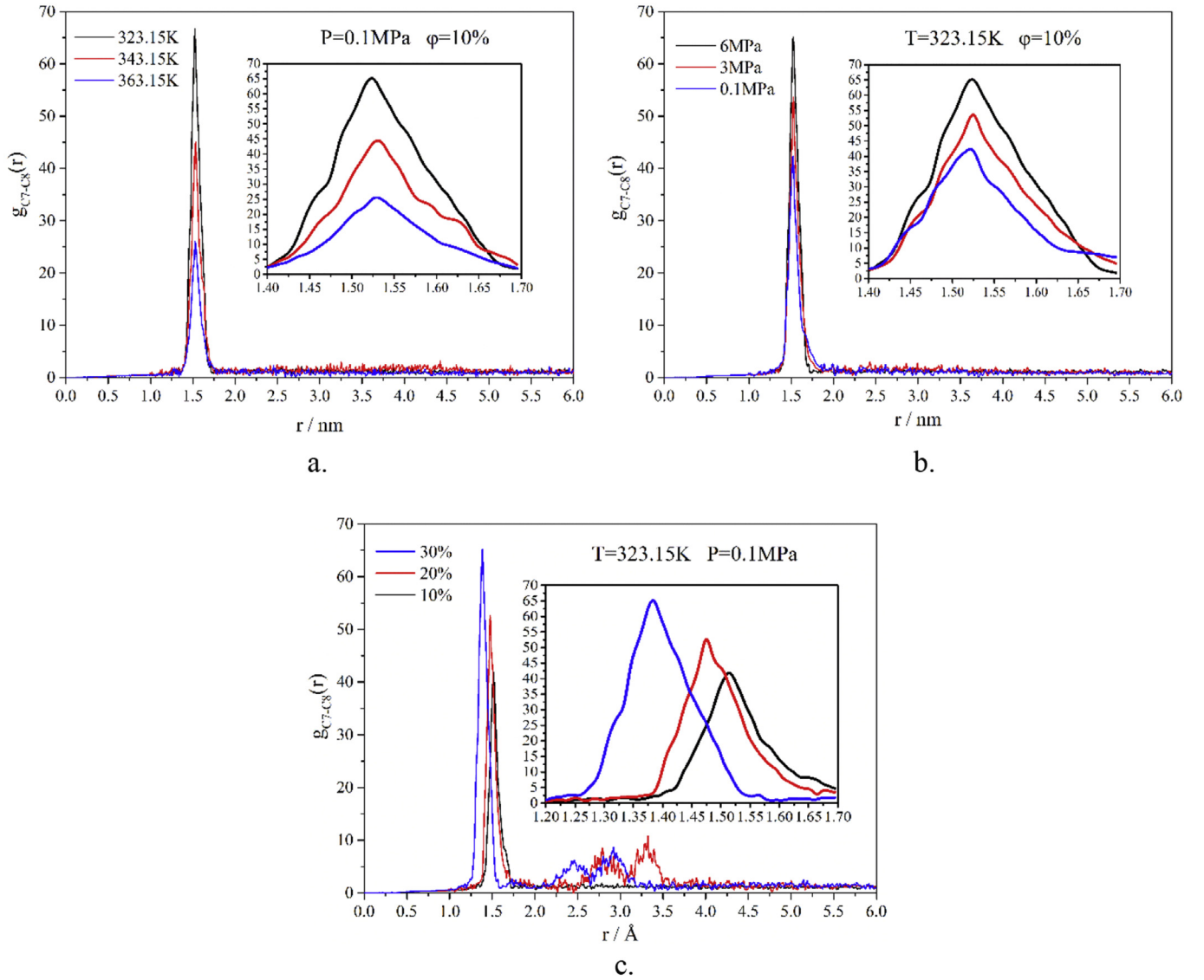


Fig. 11. Radial distribution function of wax molecules under different working conditions.

is 30%, which is lower than the inversion point (65.2%). The emulsion is still in oil-in-water state and the water is in the state of emulsified water droplets. With the increase in water cut, the bonding points of waxy crystals provided by the surface of emulsified water droplets increase, and the bonding effects between atoms in different waxy molecules are enhanced, which makes the position of peaks to the central waxy molecules appear closer. Furthermore, the wax aggregation degree becomes higher, and the concentration gradient of system will be enlarged. Therefore, the corresponding wax precipitation rate increases. The above analysis results are consistent with the conclusion obtained in reference (Duffy and Rodger, 2002): before the water cut of waxy crude oil reaches the inversion point (i.e., before the W/O emulsion translates into O/W emulsion), the wax precipitation rate will increase with the increase in water cut.

3.7. Energy analysis

For waxy crude oil multiphase system, its potential energy mainly includes bond energy, cross-term energy and non-bond energy, which consists of the following: the bond energy includes bond stretching energy, angle bending energy and dihedral angle torsion energy; the cross-term energy is the energy change caused by the coupling effect of

bonds, which mainly includes bond stretching-angle bending, angle bending-dihedral angle torsion, bond stretching-dihedral angle torsion and bond stretching-angle bending-dihedral angle torsion; and the non-bond energy mainly includes Coulomb energy, VDW (Van der Waals) energy and hydrogen bond energy. The specific energy composition can be expressed as follows:

$$E_{\text{potential}} = E_{\text{bond}} + E_{\text{non-bond}} + E_{\text{cross}} \quad (9)$$

$$E_{\text{bond}} = E_{\text{stretch}} + E_{\text{bend}} + E_{\text{torsion}} \quad (10)$$

$$E_{\text{cross}} = E_{\text{stretch-bend}} + E_{\text{bend-torsion}} + E_{\text{stretch-torsion}} + E_{\text{stretch-bend-torsion}} \quad (11)$$

$$E_{\text{non-bond}} = E_{\text{coulomb}} + E_{\text{VDW}} + E_{\text{H-bond}} \quad (12)$$

The bond energy reflects the intramolecular interaction, the cross-term energy reflects the coupling effect of bonding interaction, and the non-bond energy reflects the intermolecular interaction. In addition, the binding energy is the most intuitive physical quantity that could reflect the mutual aggregation. According to the concept of binding energy, the aggregation process is not spontaneous, which needs the power to overcome the barrier of energy (the interaction force that affects the particle aggregation). During the binding process, a certain amount of energy (binding energy) will be released from wax, and the

total energy of waxy molecular system will decrease accordingly, so the waxy molecules easily aggregate into clusters, and then form the stable crystals. In this study, the binding energy between waxy molecules is defined as:

$$E_{\text{bind,wax}} = \Sigma E_{\text{wax}} - E_{\Sigma \text{wax}} \quad (13)$$

Here, $E_{\text{bind,wax}}$ represents the binding energy between waxy molecules, ΣE_{wax} is the sum of energy when the waxy molecules exist alone, and $E_{\Sigma \text{wax}}$ is the total energy of waxy molecules after aggregation.

Sections 3.4, 3.5 and 3.6 in this study have analyzed the microscopic mechanism for the effects of different operating parameters for the dissolution, diffusion and aggregation processes. However, the affecting degree and influence mechanism of different parameters on different types of energies for waxy molecular systems is not clear. Therefore, the changing ratio of energies was defined (Formula 14). Through analyzing the changing ratios of different types of energies, we can evaluate which type of energy is more affected under different operating parameters. The influence mechanism of different working conditions on different types of energies can also be analyzed.

$$\text{Changing rate} = ((E_{\text{condition},i} - E_{\text{condition},i-1}) / E_{\text{condition},i-1}) \times 100\% \quad (14)$$

Here, $E_{\text{condition},i}$ is the arbitrary type of energy under the number i condition. $E_{\text{condition},i-1}$ is the arbitrary type of energy under the number $i-1$ condition.

According to the MD simulation results, 9 types of energies for waxy molecular systems under different temperatures, pressures and water cut were statistically calculated. Then, the corresponding changing ratios of energies were calculated, as shown in Fig. 12.

When the relaxation temperature changes from 323.15 K to 343.15 K and from 343.15 K to 363.15 K, the total potential energy decreases. Except for the increase in intramolecular bond energies, the other 5 energies all decrease, and the changing amplitudes for all energies become larger. The reasons for the above phenomena can be explained as follows: with the increase in relaxation temperature, the binding energy decreases, and the waxy molecules are difficult to aggregate into waxy clusters. As we know, the Coulomb energy, VDW energy and hydrogen bond energy all contribute to the binding interaction, and they all show the same trend as binding energy because the increase in temperature enhances the mobility of waxy molecules; the distance between molecules increases, which leads to the decrease in VDW energy. The increase in temperature also increases the electrostatic shielding effect of water on waxy molecules, and the Coulomb energy decreases accordingly. The increase in temperature will reduce the stability of hydrogen bonds between wax molecules and water molecules, and the hydrogen bond energy decreases accordingly. Moreover, due to the lower electronegativity of atoms in waxy molecules, the Coulomb interaction between waxy molecules is lower than that of VDW. In addition, the formed hydrogen bonds between water and wax are relatively weak. Therefore, among them, the VDW energy decreases the most, and the Coulomb energy takes the second place, while the changing ratio of hydrogen bond energy is the least. The reduction amplitude of cross-terms energy is almost unchanged, which indicates that the effect of temperature variation on the energy change caused by the coupling effect of intramolecular interaction is little. In terms of intramolecular bond energy, the increase in relaxation temperature will result in the severe stretching of chemical bonds between atoms. Furthermore, the amplitude of the bond angle deviating from the equilibrium and the twist rate of the dihedral angle increase, which leads to the increase in corresponding bond stretching energy, angle bending energy and dihedral angle torsion energy. The whole structure of paraffin becomes more evacuated, the distance between adjacent waxy molecules increases, and the corresponding non-bonding energy is weakened. Furthermore, the reduction of non-bonding effect also decreases the frequency of chemical bonds oscillation, so we can conclude that the functions of bond interaction and non-bond interaction

are mutually coupled. As the total potential energy is significantly affected by non-bond energy, so total potential energy shows a downward trend.

When the relaxation pressure changes from 0.1 MPa to 3 MPa and from 3 MPa to 6 MPa, the total potential energy increases. Except for the decrease in intramolecular bond energies, the other five energies all increase, and the changing amplitudes for all energies become larger. This is because, with the increase in relaxation pressure, the energies of intermolecular interaction all increase; this increase in interaction leads to the increase in binding energy because the increase in pressure weakens the mobility of waxy molecules, and the distance between them becomes closer, thus increasing the VDW energy. The increase in pressure also enhances the electrostatic shielding effect, so the Coulomb energy increases. The higher pressure also decreases the distance between waxy molecules and water molecules, which increases the hydrogen bonds between them. Among the three energies, the variation of Coulomb interaction between waxy molecules is lower than that of VDW because of the lower electronegativity for wax, and the formed hydrogen bonds are relatively weak, so the order of changing ratio is VDW energy, Coulomb energy and hydrogen bond energy. In terms of intramolecular bond energy, high pressure limits the relative motion between atoms in waxy molecules. The corresponding bond stretching energy, angle bending energy and dihedral angle torsion energy are weakened. Moreover, as the total potential energy is considerably influenced by the non-bond interaction, the total potential energy increases.

When the water cut changes from 10% to 20% and from 20% to 30%, the total potential energy increases. Except for the decrease in Coulomb energy, intramolecular bond energies and cross-term energy, the other four energies all increase, and the changing amplitudes for all energies become larger. The reason can be explained as follows. With the increase in water cut, the electrostatic shielding effect of and the distance between waxy molecules become larger. The corresponding Coulomb and VDW energies and the number of hydrogen bonds increase, and the hydrogen bond energy becomes larger. Thus, the increase in these three intermolecular interactions jointly enhance the binding energy. The decreasing amplitude of cross-term energy is relatively large because the hydrogen bonds limit the mobility of molecules, and the coupling effect of intramolecular bonds is weakened accordingly. In terms of intramolecular bond energy, the increase in water cut increases the bonding points of waxy crystals provided by the surface of emulsified water droplets, the aggregation degree becomes higher, and the formed system is more stable, which weakens the relative displacement between carbon atoms and hydrogen atoms, and the frequency of chemical bonds oscillation will be weakened accordingly. Furthermore, the total potential energy is significantly influenced by the non-bond energy, so the total potential energy shows an upward trend.

4. Conclusion

- (1) The MD model was established for characterizing the phase transition and gelation behavior of waxy molecules in crude oil multiphase system. The simulation results were basically in agreement with the experimental results of Dutour's et al. (2002), and the relative error was less than 5%, which illustrated that the results simulated by the MD model were reliable.
- (2) The microscopic processes of waxy crystals in crude oil multiphase systems can be described as follows: first, the initial large waxy spherical crystal dissolves in the multiphase system; then, they diffuse continuously and form the irregular distribution. Furthermore, the waxy molecules are aggregated into crystals with smaller volume and formed the gelation structure; finally, the waxy crystals will precipitate driven by concentration gradient.
- (3) The water and asphaltene molecules have the effect on the processes of dissolution, diffusion and aggregation for waxy molecules.

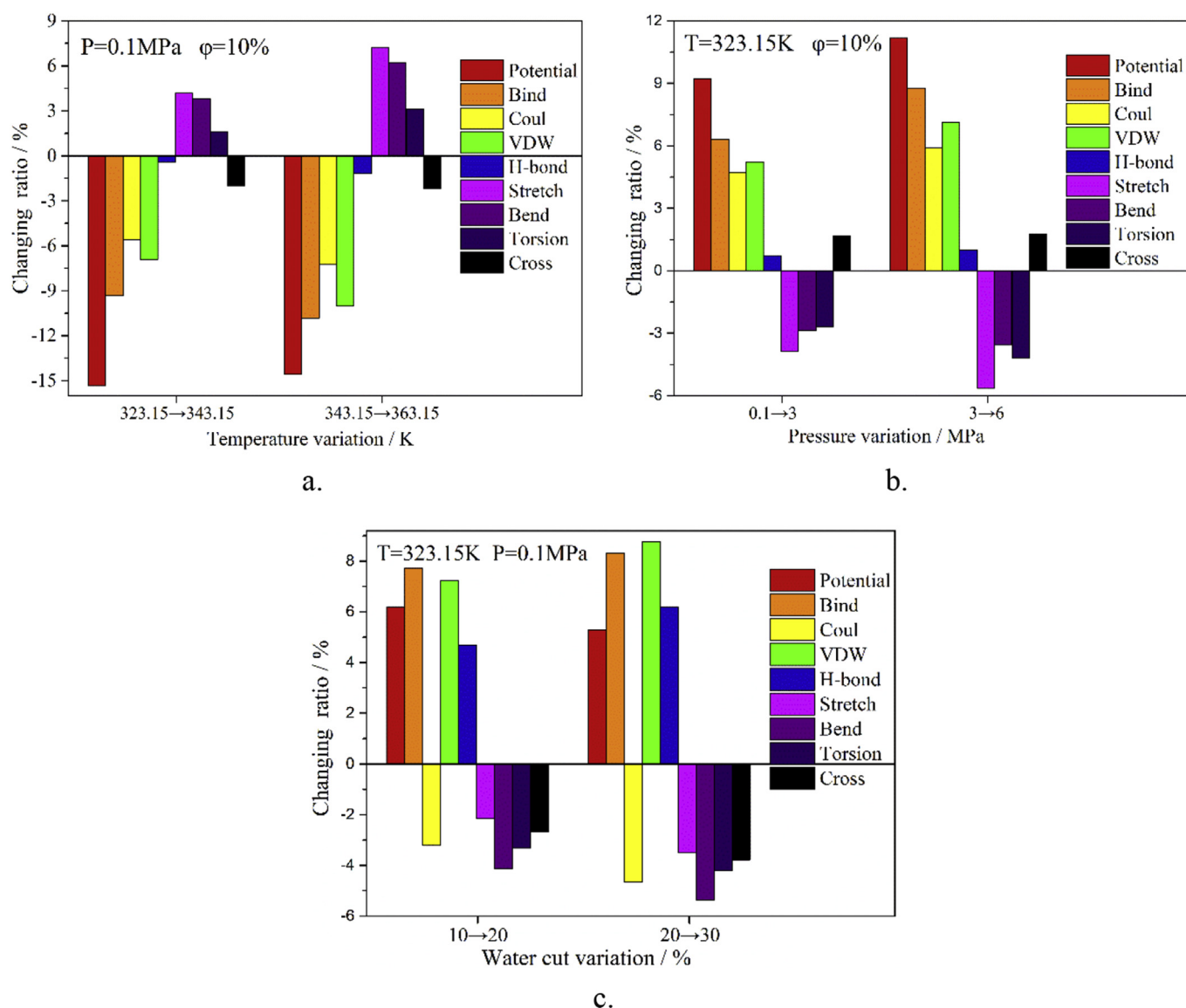


Fig. 12. Changing ratio of energies for wax molecules under the variation of parameters.

For water, the hydrogen bond with the structure of $\text{C}-\text{H}\cdots\text{O}$ will form, which affects the entire phase transition and gelation processes. For asphaltene, the wax is easier to dissolve in the multi-phase system according to the principle of similar dissolve mutually theory.

- (4) The influence mechanism of different operation parameters on the dissolution, diffusion and aggregation processes was analyzed. With the increase in temperature, the dissolution and diffusion abilities of waxy molecules increase, the aggregation degree decreases, which reduces the waxy gelation degree. With the increase in pressure, the dissolution and diffusion abilities increase, the aggregation degree of waxy crystals increases, which makes it easier to form waxy crystals with larger amount and larger volume, as such, the degree of wax gelation increases. With the increase in water cut, the dissolution abilities of waxy crystals are enhanced, the diffusion ability decreases, the aggregation degree increased, the wax gelation degree increases accordingly.

Acknowledgments

This work was financially supported by the National Natural Science Foundation of China (51534004) and the Northeast Petroleum

University Innovation Foundation for Postgraduate (JYCX_CX02_2018).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.petrol.2019.04.059>.

References

- Adams, J.J., Tort, F., Schabron, J.F., Loveridge, J.L., Rovani, J., Baig, K., 2018. Evaluation of wax inhibitor performance through various techniques. *Energy Fuel*. 32 (12), 12151–12165. <https://doi.org/10.1021/acs.energyfuels.8b02693>.
- Boek, E.S., Yakovlev, D.S., Headen, T.F., 2009. Quantitative molecular representation of asphaltenes and molecular dynamics simulation of their aggregation. *Energy Fuel*. 23 (3), 1209–1219. <https://doi.org/10.1021/ef800876b>.
- Cao, W., Xie, K., Lu, X., Liu, Y., Zhang, Y., 2019. Effect of profile-control oil-displacement agent on increasing oil recovery and its mechanism. *Fuel* 237, 1151–1160. <https://doi.org/10.1016/j.fuel.2018.10.089>.
- Chen, X., Hou, L., Li, W., Li, S., Chen, Y., 2017. Molecular dynamics simulation of magnetic field influence on waxy crude oil. *J. Mol. Liq.* 249 (1), 1052–1059. <https://doi.org/10.1016/j.molliq.2017.11.101>.
- Cheng, Q., Pan, C., Zhao, Y., Liu, Y., Sun, W., Gan, Y., 2017. Phenomenological study on heat and mass coupling mechanism of waxy crude oil pipeline transport process. *J. Dispersion Sci. Technol.* 38 (9), 1276–1284. <https://doi.org/10.1080/01932691.2016.1234383>.
- Duffy, D.M., Rodger, P.M., 2002. Modeling the activity of wax inhibitors: a case study of

- poly (octadecyl acrylate). *J. Phys. Chem. B* 106 (43), 11210–11217. <https://doi.org/10.1021/jp026501j>.
- Duffy, D.M., Moon, C., Rodger, P.M., 2004. Computer-assisted design of oil additives: hydrate and wax inhibitors. *Mol. Phys.* 102 (2), 203–210. <https://doi.org/10.1080/00268970310001648717>.
- Dutour, S., Lagourette, B., Daridon, J.L., 2002. High-pressure speed of sound, density and compressibility of heavy normal paraffins: $C_{28}H_{58}$, and $C_{36}H_{74}$. *J. Chem. Thermodyn.* 34 (4), 475–484. <https://doi.org/10.1006/jcht.2001.0922>.
- Feller, S.E., Zhang, Y., Pastor, R.W., 1995. Constant pressure molecular dynamics simulation: the Langevin piston method. *J. Chem. Phys.* 103 (11), 4613–4621. <https://doi.org/10.1063/1.470648>.
- Gan, Y.F., Cheng, Q.L., Sun, W., Gao, W., Liu, X., Liu, Y., 2018. The stability criterion model and stability analysis of waxy crude oil pipeline transportation system based on excess entropy production. *J. Therm. Sci.* 27 (6), 1–14. <https://doi.org/10.1007/s11630-018-1048-6>.
- Gao, F., Xu, Z., Liu, G., Yuan, S., 2014. Molecular dynamics simulation: the behavior of asphaltene in crude oil and at the oil/water interface. *Energy Fuel*. 28 (12), 7368–7376. <https://doi.org/10.1021/ef5020428>.
- Hayduk, W., Minhas, B.S., 1982. Correlations for prediction of molecular diffusivities in liquids. *Can. J. Chem. Eng.* 60 (2), 295–299. <https://doi.org/10.1002/cjce.5450600213>.
- Jang, Y.H., Blanco, M., Creek, J., Tang, Y., Goddard, W.A., 2007. Wax inhibition by comb-like polymers: support of the incorporation-perturbation mechanism from molecular dynamics simulations. *J. Phys. Chem. B* 111 (46), 13173–13179. <https://doi.org/10.1021/jp072792q>.
- Jorgensen, W.L., D., S.M., Tiradorives, J., 1996. Development and testing of the OPLS All-Atom force field on conformational energetics and properties of organic liquids. *J. Am. Chem. Soc.* 118 (45), 11225–11236. <https://doi.org/10.1021/ja9621760>.
- Kahn, K., Bruice, T.C., 2002. Parameterization of OPLS-AA force field for the conformational analysis of macrocyclic polyketides. *J. Comput. Chem.* 23 (10), 977–996. <https://doi.org/10.1002/jcc.10051>.
- Kané, M., Djabourov, M., Volle, J.L., Lechaire, J.P., Frebourg, G., 2003. Morphology of paraffin crystals in waxy crude oils cooled in quiescent conditions and under flow. *Fuel* 82 (2), 127–135. [https://doi.org/10.1016/S0016-2361\(02\)00222-3](https://doi.org/10.1016/S0016-2361(02)00222-3).
- Kriz, P., Andersen, S.I., 2004. Effect of asphaltenes on crude oil wax crystallization. *Energy Fuel*. 19 (3), 948–953. <https://doi.org/10.1021/ef049819e>.
- Li, C.X., Cai, J.Y., Cheng, L., Yang, F., Zhang, H., Zhang, Y., 2016. Stratification phenomenon and laws of wax deposits of waxy oil triggered by asphaltene addition. *CIE J.* 67 (6), 2426–2432. <https://doi.org/10.11949/j.issn.0438-1157.20151723>.
- Liu, Y., Chen, G., 1999. Optimal parameters design of oilfield surface pipeline systems using fuzzy models. *Inf. Sci.* 120 (1–4), 13–21. [https://doi.org/10.1016/S0020-0255\(99\)00059-6](https://doi.org/10.1016/S0020-0255(99)00059-6).
- Liu, Y., Wang, Z.H., Zhuge, X., Zhao, S., Pang, R., 2012. Study of deposition behavior in small-diameter gathering pipelines for water-cut oil. *Chem. Technol. Fuels Oils* 48 (5), 386–392. <https://doi.org/10.1007/s10553-012-0385-5>.
- Liu, Y., Pan, C., Cheng, Q., Wang, B., Wang, X., Gan, Y., 2018. Wax deposition rate model for heat and mass coupling of piped waxy crude oil based on non-equilibrium thermodynamics. *J. Dispersion Sci. Technol.* 39 (2), 259–269. <https://doi.org/10.1080/01932691.2017.1312432>.
- Luo, H., Wang, R., Fan, W., Li, Z., Ma, T., Nan, G., 2005. Calculation of solubility parameters of supercritical CO_2 by molecular dynamics simulation. *Acta Petrol. Sin.* 31 (1), 78–83. <https://doi.org/10.3969/j.issn.1001-8719.2015.01.013>.
- Martín-Martínez, F.J., Fini, E.H., Buehler, M.J., 2015. Molecular asphaltene models based on Clar sextet theory. *RSC Adv.* 5 (1), 753–759. <https://doi.org/10.1039/C4RA05694A>.
- Migdisov, A.A., Bychkov, A.Y., Williams-Jones, A.E., van Hinsberg, V.J., 2014. A predictive model for the transport of copper by HCl-bearing water vapour in ore-forming magmatic-hydrothermal systems: implications for copper porphyry ore formation. *Geochim. Cosmochim. Acta* 129 (4), 33–53. <https://doi.org/10.1016/j.gca.2013.12.024>.
- Morishita, T., 2010. From Nosé–Hoover chain to Nosé–Hoover network: design of non-Hamiltonian equations of motion for molecular-dynamics with multiple thermostats. *Mol. Phys.* 108 (10), 1337–1347. <https://doi.org/10.1080/00268971003689923>.
- Panacharoensawad, E., Sarica, C., 2012. Experimental study of single-phase and two-phase water-in-crude-oil dispersed flow wax deposition in a mini pilot-scale flow loop. *Energy Fuel*. 27 (9), 5036–5053. <https://doi.org/10.1021/ef400275h>.
- Piroozian, A., Hemmati, M., Ismail, I., Manan, M.A., Bayat, A.E., Mohsin, R., 2016. Effect of emulsified water on the wax appearance temperature of water-in-waxy-crude-oil emulsions. *Thermochim. Acta* 637, 132–142. <https://doi.org/10.1016/j.tca.2016.05.014>.
- Quan, Q., Gong, J., Wang, W., Gao, G., 2015. Study on the aging and critical carbon number of wax deposition with temperature for crude oils. *J. Pet. Sci. Eng.* 130 (6), 1–5. <https://doi.org/10.1016/j.petrol.2015.03.026>.
- Raghavendra, B., Arunan, E., 2008. Hydrogen bonding with a hydrogen bond: the methane-water complex and the penta-coordinate carbon. *Chem. Phys. Lett.* 467 (1), 37–40. <https://doi.org/10.1016/j.cplett.2008.11.009>.
- Samieadel, A., Oldham, D., Fini, E.H., 2017. Multi-scale characterization of the effect of wax on intermolecular interactions in asphalt binder. *Constr. Build. Mater.* 157, 1163–1172. <https://doi.org/10.1016/j.conbuildmat.2017.09.188>.
- Samieadel, A., Oldham, D., Fini, E.H., 2018. Investigating molecular conformation and packing of oxidized asphaltene molecules in presence of paraffin wax. *Fuel* 220, 503–512. <https://doi.org/10.1016/j.fuel.2018.02.031>.
- Singh, P., Venkatesan, R., Fogler, H.S., Nagarajan, N., 2000. Formation and aging of incipient thin film wax-oil gels. *AIChE J.* 46 (5), 1059–1074. <https://doi.org/10.1002/aic.690460517>.
- Szczeniński, G., Chalasiński, G., Cybulski, S.M., Cieplak, P., 1993. Ab initio study of the potential energy surface of CH_4-H_2O . *J. Chem. Phys.* 98 (4), 3078–3089. <https://doi.org/10.1063/1.465050>.
- Taheri-Shakib, J., Rajabi-Kochi, M., Kazemzadeh, E., Naderi, H., Shekarifard, A., 2018a. A comprehensive study of the impact of wax compositions on the wax appearance temperature (WAT) of some Iranian crude oils: an experimental investigation. *J. Pet. Sci. Eng.* 165 (6), 67–80. <https://doi.org/10.1016/j.petrol.2018.02.002>.
- Taheri-Shakib, J., Shekarifard, A., Naderi, H., 2018b. Characterization of the wax precipitation in Iranian crude oil based on Wax Appearance Temperature (WAT): Part 1. The influence of electromagnetic waves. *J. Pet. Sci. Eng.* 161 (2), 530–540. <https://doi.org/10.1016/j.petrol.2017.12.012>.
- Tang, J., Qu, Z., Luo, J., He, L., Wang, P., Zheng, P., Tang, X., Pei, Y., Ding, B., Peng, B., Huang, Y., 2018. Molecular dynamics simulations of the oil-detachment from the hydroxylated silica surface: effects of surfactants, electrostatic interactions and water flows on the water molecular channel formation. *J. Phys. Chem. B* 122 (6), 1905–1918. <https://doi.org/10.1021/acs.jpcc.7b09716>.
- Wang, S., Feng, Q., Zha, M., Lu, S., Qin, Y., Xia, T., Zhang, C., 2015. Molecular dynamics simulation of liquid alkane occurrence state in pores and slits of shale organic matter. *Petrol. Explor. Dev.* 42 (6), 844–851. [https://doi.org/10.1016/S1876-3804\(15\)30081-1](https://doi.org/10.1016/S1876-3804(15)30081-1).
- Wu, C., Zhang, J., Li, W., Wu, N., 2005. Molecular dynamics simulation guiding the improvement of EVA-type pour point depressant. *Fuel* 84 (16), 2039–2047. <https://doi.org/10.1016/j.fuel.2004.12.009>.
- Xie, K., Lu, X., Pan, H., Han, D., Hu, G., Zhang, J., Zhang, B., Cao, B., 2018. Analysis of dynamic imbibition effect of surfactant in microcracks of reservoir at high temperature and low permeability. *SPE Prod. Oper.* 33 (3), 596–606. <https://doi.org/10.2118/189970-PA>.
- Yao, B., Li, C., Yang, F., Sjöblom, J., Zhang, Y., Norrman, J., Paso, K., Xiao, Z., 2016. Organically modified nano-clay facilitates pour point depressing activity of polyoctadecylacrylate. *Fuel* 166, 96–105. <https://doi.org/10.1016/j.fuel.2015.10.114>.
- Yao, B., Li, C., Yang, F., Zhang, X., Mu, Z., Sun, G., Zhao, Y., 2018. Ethylene-vinyl acetate copolymer and resin-stabilized asphaltenes synergistically improve the flow behavior of model waxy oils. 1. effect of wax content and the synergistic mechanism. *Energy Fuel*. 32 (2), 1567–1578. <https://doi.org/10.1021/acs.energyfuels.7b03657>.
- Zhang, Z., Duan, Z., 2005. Prediction of the PVT properties of water over wide range of temperatures and pressures from molecular dynamics simulation. *Phys. Earth Planet. In.* 149 (3), 335–354. <https://doi.org/10.1016/j.pepi.2004.11.003>.
- Zhang, J., Zhang, M., Wan, J., Li, W., 2008. Theoretical study of the prohibited mechanism for ethylene/vinyl acetate co-polymers to the wax crystal growth. *J. Phys. Chem. B* 112 (1), 36–43. <https://doi.org/10.1021/jp073052k>.
- Zheng, S., Fogler, H.S., 2014. Fundamental investigation of wax diffusion characteristics in water-in-oil emulsion. *Ind. Eng. Chem. Res.* 54 (16), 4420–4428. <https://doi.org/10.1021/ie501955e>.
- Zhou, Q., Lu, X., Liu, X., Zhang, L., He, H., Zhu, J., Yuan, P., 2011. Hydration of methane intercalated in Na-smectites with distinct layer charge: insights from molecular simulations. *J. Colloid Interface Sci.* 355 (1), 237–242. <https://doi.org/10.1016/j.jcis.2010.11.054>.